

Detecting and removing malware using VirusTotal integration

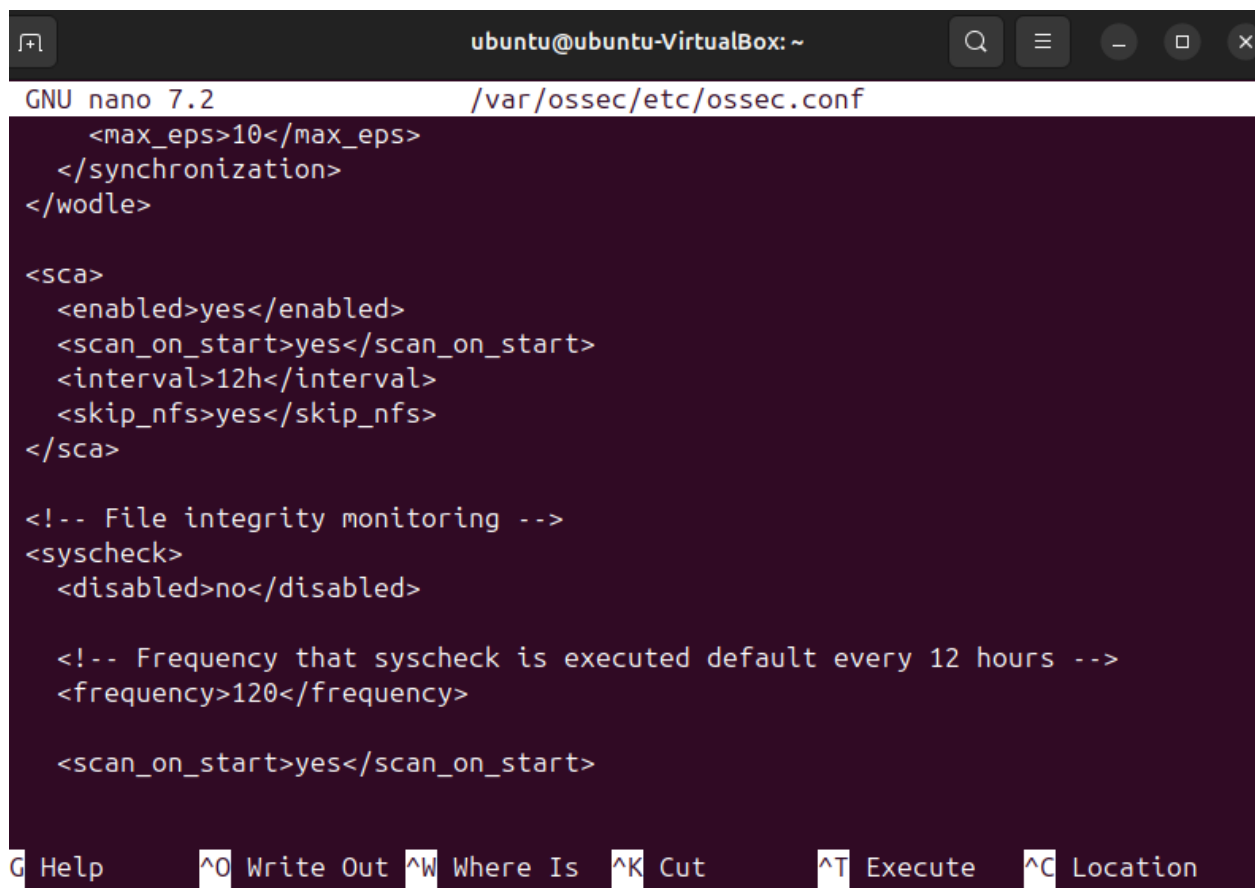
- Google Wazuh Documentation – Proof of concept guide - Detecting and removing malware using VirusTotal integration

Follow the instructions which starts with ubuntu

Ubuntu Endpoint

1. [Sudo nano/var/ossec/etc/ossec.conf](#). Make sure that <disabled> is set to no. This is to monitor for directory changes by Wazuh FIM.

Changed frequency to 120



The screenshot shows a terminal window titled 'ubuntu@ubuntu-VirtualBox: ~'. The nano editor is open to the file '/var/ossec/etc/ossec.conf'. The visible content of the file is as follows:

```
GNU nano 7.2 /var/ossec/etc/ossec.conf
  <max_eps>10</max_eps>
</synchronization>
</wodle>

<sca>
  <enabled>yes</enabled>
  <scan_on_start>yes</scan_on_start>
  <interval>12h</interval>
  <skip_nfs>yes</skip_nfs>
</sca>

<!-- File integrity monitoring -->
<syscheck>
  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>120</frequency>

  <scan_on_start>yes</scan_on_start>
```

At the bottom of the terminal, a status bar shows various keyboard shortcuts: 'G Help', '^O Write Out', '^W Where Is', '^K Cut', '^T Execute', and '^C Location'.

2. Add [<directories realtime="yes">/root</directories>](#) to monitor root directories

```
ubuntu@ubuntu-VirtualBox: ~  
GNU nano 7.2 /var/ossec/etc/ossec.conf  
  <enabled>yes</enabled>  
  <scan_on_start>yes</scan_on_start>  
  <interval>12h</interval>  
  <skip_nfs>yes</skip_nfs>  
</sca>  
  
  <!-- File integrity monitoring -->  
  <syscheck>  
    <disabled>no</disabled>  
  
    <!-- Frequency that syscheck is executed default every 12 hours -->  
    <frequency>120</frequency>  
  
    <scan_on_start>yes</scan_on_start>  
  
    <!-- Directories to check (perform all possible verifications) -->  
    <directories>/etc,/usr/bin,/usr/sbin</directories>  
    <directories>/bin,/sbin,/boot</directories>  
    <directories realtime="yes">/root</directories>  
    <!-- Files/directories to ignore -->  
  
G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  
X Exit      ^R Read File ^\ Replace  ^U Paste     ^J Justify   ^_ Go To Line
```

3. install jq, a utility that processes JSON input from the active response script.

`sudo apt update`

`sudo apt -y install jq`

```
ubuntu@ubuntu-VirtualBox: ~  
et:23 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages  
1,188 kB]  
et:24 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en  
229 kB]  
et:25 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Component  
[74.2 kB]  
et:26 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Package  
[39.3 kB]  
et:27 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-e  
[8,564 B]  
etched 15.9 MB in 2min 23s (111 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
85 packages can be upgraded. Run 'apt list --upgradable' to see them.  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
jq is already the newest version (1.7.1-3ubuntu0.24.04.2).  
The following package was automatically installed and is no longer required:  
  libwoff1  
Use 'sudo apt autoremove' to remove it.  
0 upgraded, 0 newly installed, 0 to remove and 185 not upgraded.  
ubuntu@ubuntu-VirtualBox:~$
```

4. Paste sudo nano /var/ossec/active-response/bin/remove-threat.sh active and paste the below to response script to remove malicious files from the endpoint.

```
GNU nano 7.2 /var/ossec/active-response/bin/remove-threat.sh
#!/bin/bash

LOCAL=`dirname $0`;
cd $LOCAL
cd ../

PWD=`pwd`

read INPUT_JSON
FILENAME=$(echo $INPUT_JSON | jq -r .parameters.alert.data.virustotal.source.file)
COMMAND=$(echo $INPUT_JSON | jq -r .command)
LOG_FILE="${PWD}/../logs/active-responses.log"

#----- Analyze command -----#
if [ ${COMMAND} = "add" ]
then
# Send control message to execd
printf '{"version":1,"origin":{"name":"remove-threat","module":"active-respons>

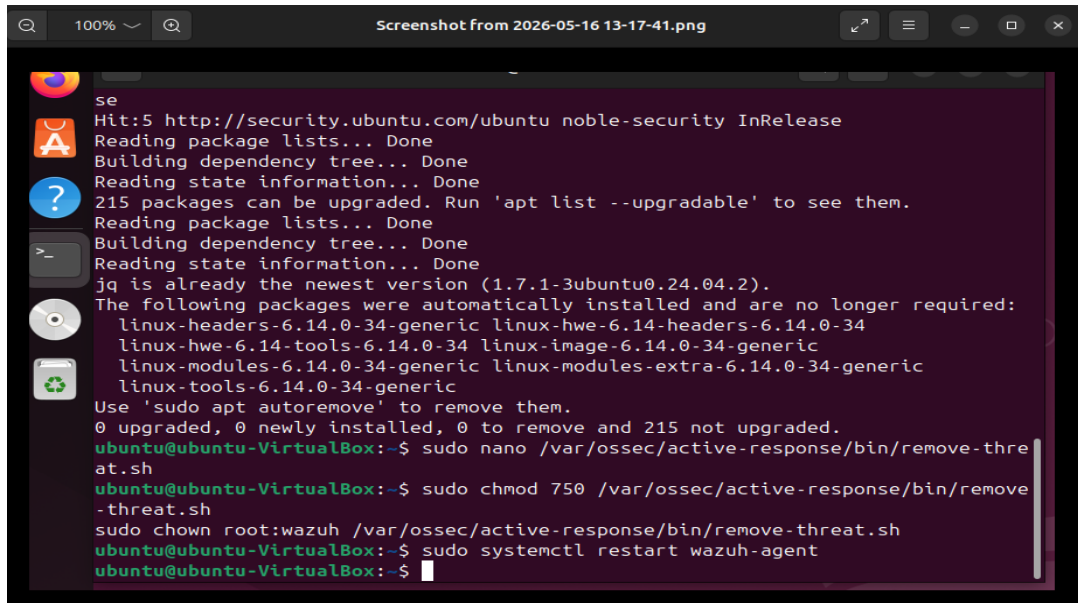
read RESPONSE
[ Read 37 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

5.Changing file ownership and Permission

```
s [39.3 kB]
Get:27 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-e
n [8,564 B]
Fetched 15.9 MB in 2min 23s (111 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
185 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
jq is already the newest version (1.7.1-3ubuntu0.24.04.2).
The following package was automatically installed and is no longer required:
  libwoff1
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 185 not upgraded.
ubuntu@ubuntu-VirtualBox:~$ sudo nano ^[[200~/var/ossec/active-response/bin/remo
ve-threat.sh~
ubuntu@ubuntu-VirtualBox:~$ sudo nano /var/ossec/active-response/bin/remove-thre
at.sh
ubuntu@ubuntu-VirtualBox:~$ sudo chmod 750 /var/ossec/active-response/bin/remove
-threat.sh
sudo chown root:wazuh /var/ossec/active-response/bin/remove-threat.sh
ubuntu@ubuntu-VirtualBox:~$
```

- Restart Wazuh to apply changes.

```
sudo systemctl restart wazuh-agent
```



The screenshot shows a terminal window titled "Screenshot from 2026-05-16 13-17-41.png". The terminal output shows the following commands and their results:

```
se
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
215 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
jq is already the newest version (1.7.1-3ubuntu0.24.04.2).
The following packages were automatically installed and are no longer required:
  linux-headers-6.14.0-34-generic linux-hwe-6.14-headers-6.14.0-34
  linux-hwe-6.14-tools-6.14.0-34 linux-image-6.14.0-34-generic
  linux-modules-6.14.0-34-generic linux-modules-extra-6.14.0-34-generic
  linux-tools-6.14.0-34-generic
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 215 not upgraded.
ubuntu@ubuntu-VirtualBox:~$ sudo nano /var/ossec/active-response/bin/remove-threat.sh
ubuntu@ubuntu-VirtualBox:~$ sudo chmod 750 /var/ossec/active-response/bin/remove-threat.sh
sudo chown root:wazuh /var/ossec/active-response/bin/remove-threat.sh
ubuntu@ubuntu-VirtualBox:~$ sudo systemctl restart wazuh-agent
ubuntu@ubuntu-VirtualBox:~$
```

WAZUH SERVER

Since you can not paste on wazuh, we enable ssh

On chatgpt, typed "How do I confirm the distribution of my linux system"

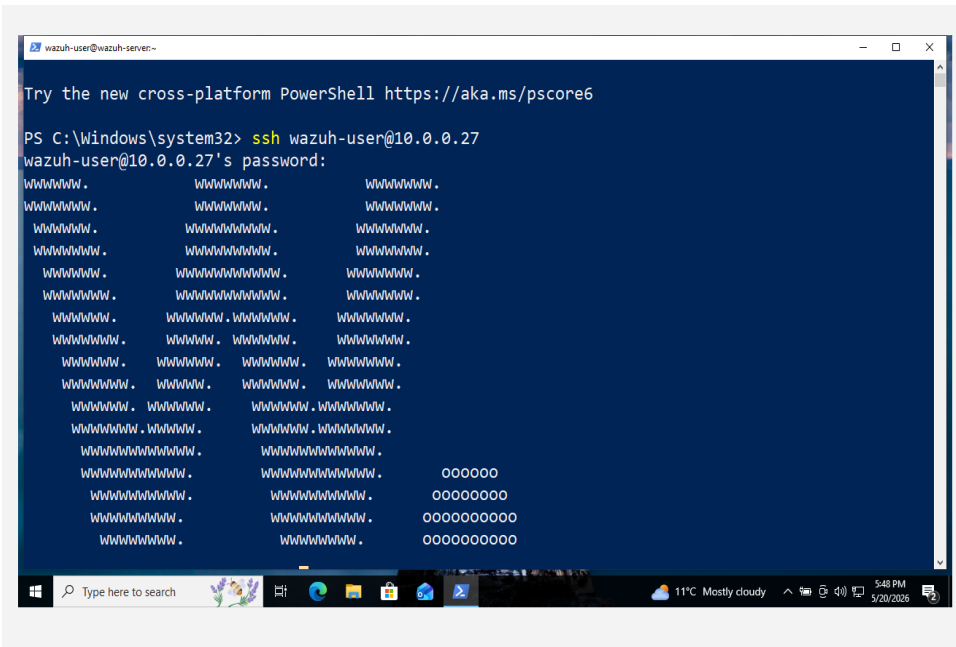
`cat /etc/os-release` – typed into wazuh which indicated we were using amazon

Types ssh for amazon - `Sudo systemctl status sshd`

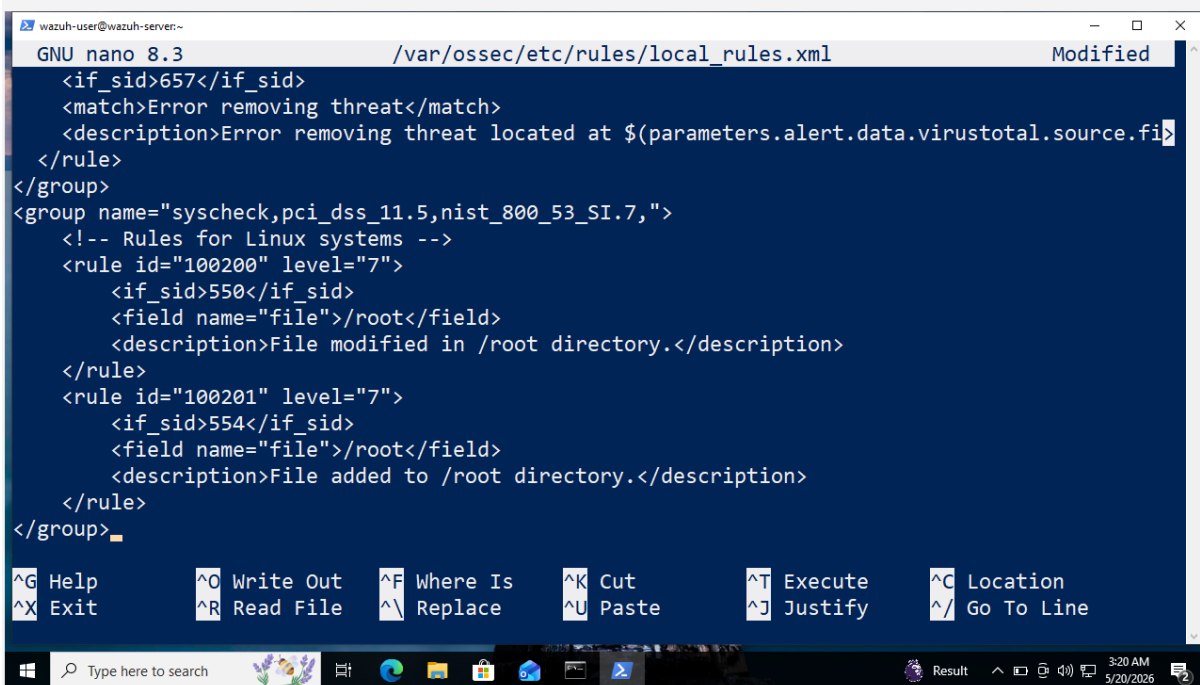
```
NAME="Amazon Linux"
VERSION="2023"
ID="amzn"
ID_LIKE="fedora"
VERSION_ID="2023"
PLATFORM_ID="platform:al2023"
PRETTY_NAME="Amazon Linux 2023.11.20260413"
ANSI_COLOR="0:33"
CPE_NAME="cpe:2.3:o:amazon:amazon_linux:2023"
HOME_URL="https://aws.amazon.com/linux/amazon-linux-2023/"
DOCUMENTATION_URL="https://docs.aws.amazon.com/linux/"
SUPPORT_URL="https://aws.amazon.com/premiumsupport/"
BUG_REPORT_URL="https://github.com/amazonlinux/amazon-linux-2023"
VENDOR_NAME="AWS"
VENDOR_URL="https://aws.amazon.com/"
SUPPORT_END="2029-06-30"
[wazuh-user@wazuh-server ~]$ sudo systemctl status sshd
■ sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-05-17 00:07:49 UTC; 48min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 1996 (sshd)
     Tasks: 1 (limit: 4647)
    Memory: 2.8M
       CPU: 66ms
   CGroup: /system.slice/ssh.service
           └─1996 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

May 17 00:07:49 wazuh-server sshd[1996]: Server listening on 0.0.0.0 port 22.
May 17 00:07:49 wazuh-server sshd[1996]: Server listening on :: port 22.
[wazuh-user@wazuh-server ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1f:15:42 brd ff:ff:ff:ff:ff:ff
    altname enp0s17
    inet 10.0.0.27/24 metric 1024 brd 10.0.0.255 scope global dynamic eth0
        valid_lft 169865sec preferred_lft 169865sec
    inet6 2604:3409:147c:31e0:aa04:128 scope global dynamic noprefixroute
        valid_lft 169883sec preferred_lft 169883sec
    inet6 2604:3409:147c:31e0:a09:271f:fe1f:1542/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 172798sec preferred_lft 172798sec
    inet6 fe80::a09:271f:fe1f:1542/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[wazuh-user@wazuh-server ~]$
```

To get wazuh on Windows, type `ssh wazuh-user@10.0.0.27 -wazuh ip`



1. `sudo nano /var/ossec/etc/rules/local_rules.xml` and paste below



2. sudo nano /var/ossec/etc/ossec.conf and replace API key gotten from virus integration account.

```
GNU nano 8.3 /var/ossec/etc/ossec.conf
<timeout_allowed>no</timeout_allowed>
</command>

<active-response>
  <disabled>no</disabled>
  <command>remove-threat</command>
  <location>local</location>
  <rules_id>87105</rules_id>
</active-response>
</ossec_config>
<ossec_config>
  <integration>
    <name>virustotal</name>
    <api_key>ab268c19855a4869bb6790732c2ee44cbdb53aba299eff3974a45cf08c1973b0</api_key> <!-- R
    <group>syscheck</group>
    <alert_format>json</alert_format>
  </integration>
</ossec_config>
```

Help Write Out Where Is Cut Execute Location
Exit Read File Replace Paste Justify Go To Line

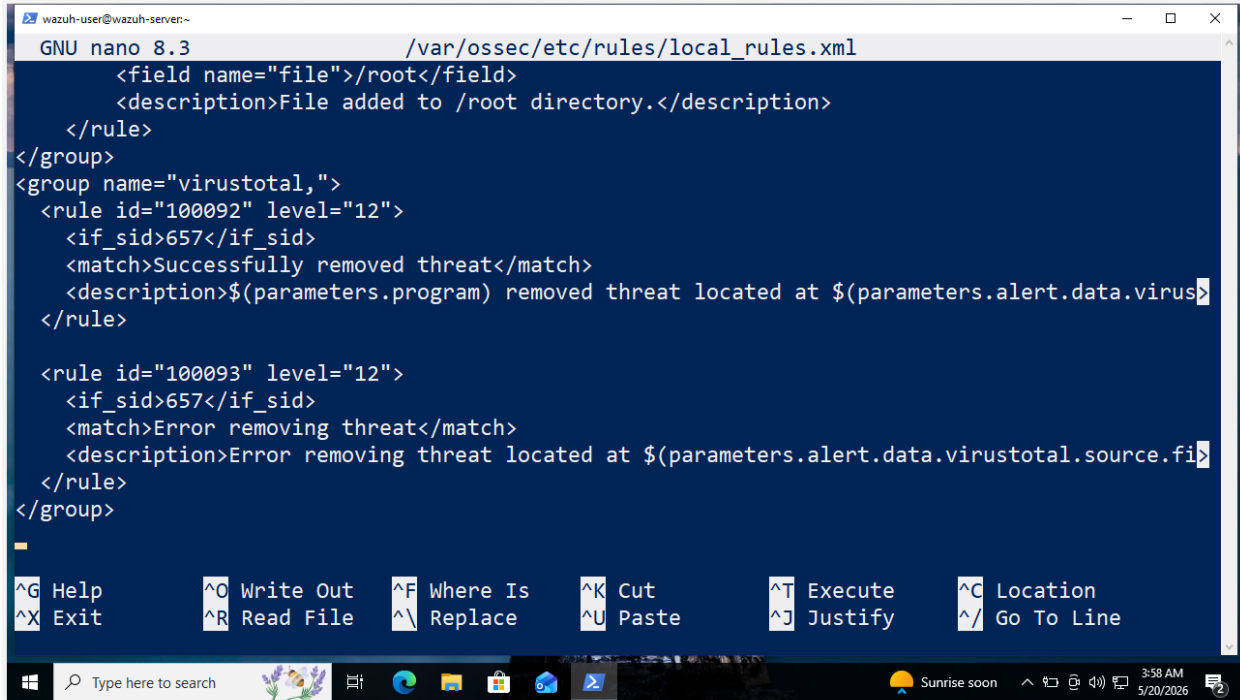
3. Sudo nano /var/ossec/etc/ossec.conf. paste in the below to flag malicious file

```
GNU nano 8.3 /var/ossec/etc/ossec.conf
<alert_format>json</alert_format>
</integration>
</ossec_config>
<ossec_config>
  <command>
    <name>remove-threat</name>
    <executable>remove-threat.exe</executable>
    <timeout_allowed>no</timeout_allowed>
  </command>

  <active-response>
    <disabled>no</disabled>
    <command>remove-threat</command>
    <location>local</location>
    <rules_id>87105</rules_id>
  </active-response>
</ossec_config>
```

Help Write Out Where Is Cut Execute Location
Exit Read File Replace Paste Justify Go To Line

4. `sudo nano /var/ossec/etc/rules/local_rules.xml` – Alert about active erespone result.

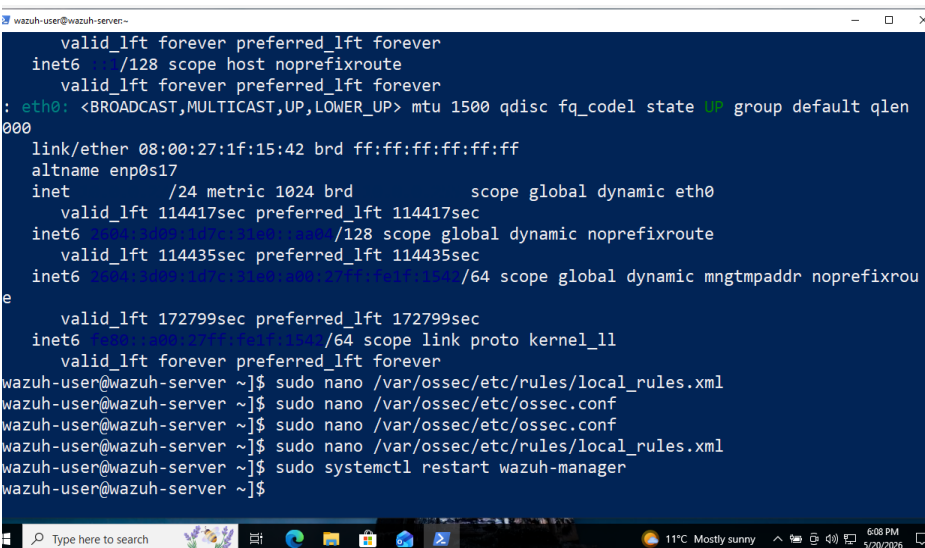


```
GNU nano 8.3 /var/ossec/etc/rules/local_rules.xml
<field name="file">/root</field>
<description>File added to /root directory.</description>
</rule>
</group>
<group name="virustotal,">
<rule id="100092" level="12">
<if_sid>657</if_sid>
<match>Successfully removed threat</match>
<description>$(parameters.program) removed threat located at $(parameters.alert.data.virus
</rule>

<rule id="100093" level="12">
<if_sid>657</if_sid>
<match>Error removing threat</match>
<description>Error removing threat located at $(parameters.alert.data.virustotal.source.fi
</rule>
</group>
```

5. Restart to apply configuration

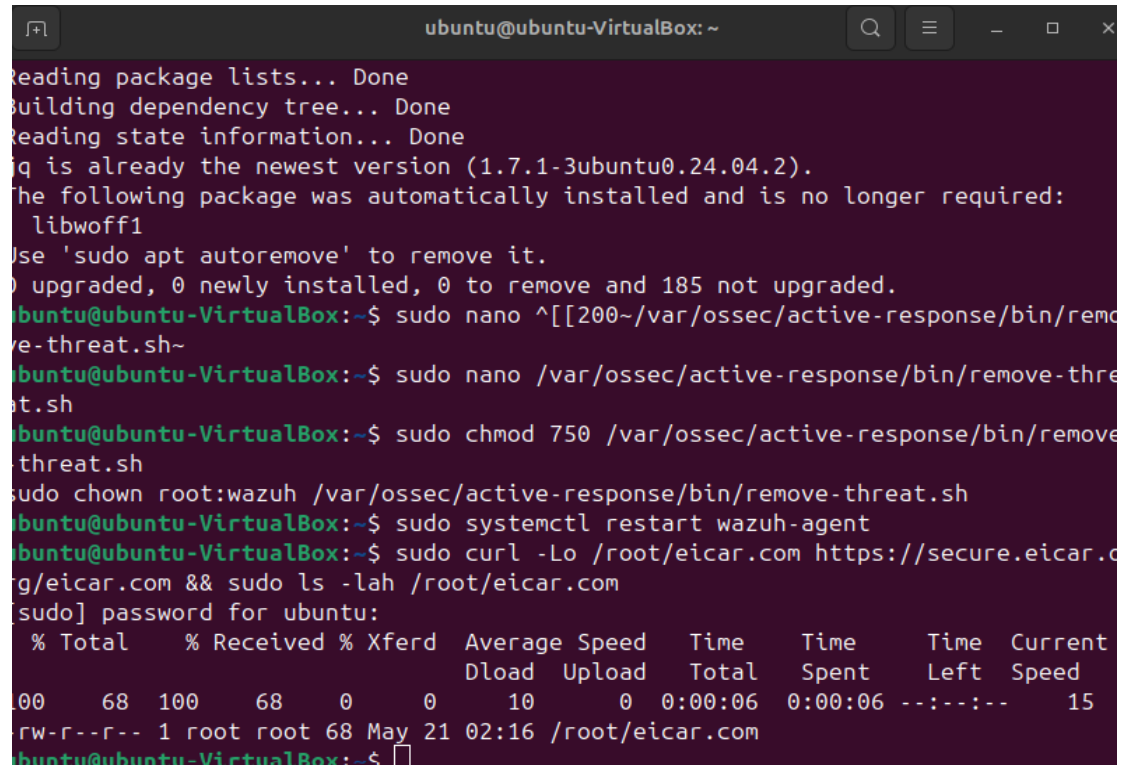
`sudo systemctl restart wazuh-manager`



```
valid_lft forever preferred_lft forever
inet6 /128 scope host noprefixroute
valid_lft forever preferred_lft forever
: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen
000
link/ether 08:00:27:1f:15:42 brd ff:ff:ff:ff:ff:ff
altname enp0s17
inet /24 metric 1024 brd scope global dynamic eth0
valid_lft 114417sec preferred_lft 114417sec
inet6 /128 scope global dynamic noprefixroute
valid_lft 114435sec preferred_lft 114435sec
inet6 /64 scope global dynamic mngtmpaddr noprefixrou
e
valid_lft 172799sec preferred_lft 172799sec
inet6 /64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
wazuh-user@wazuh-server ~]$ sudo nano /var/ossec/etc/rules/local_rules.xml
wazuh-user@wazuh-server ~]$ sudo nano /var/ossec/etc/ossec.conf
wazuh-user@wazuh-server ~]$ sudo nano /var/ossec/etc/ossec.conf
wazuh-user@wazuh-server ~]$ sudo nano /var/ossec/etc/rules/local_rules.xml
wazuh-user@wazuh-server ~]$ sudo systemctl restart wazuh-manager
wazuh-user@wazuh-server ~]$
```


ATTACK EMULATION

```
sudo curl -Lo /root/eicar.com https://secure.eicar.org/eicar.com && sudo ls -lah /root/eicar.com
```

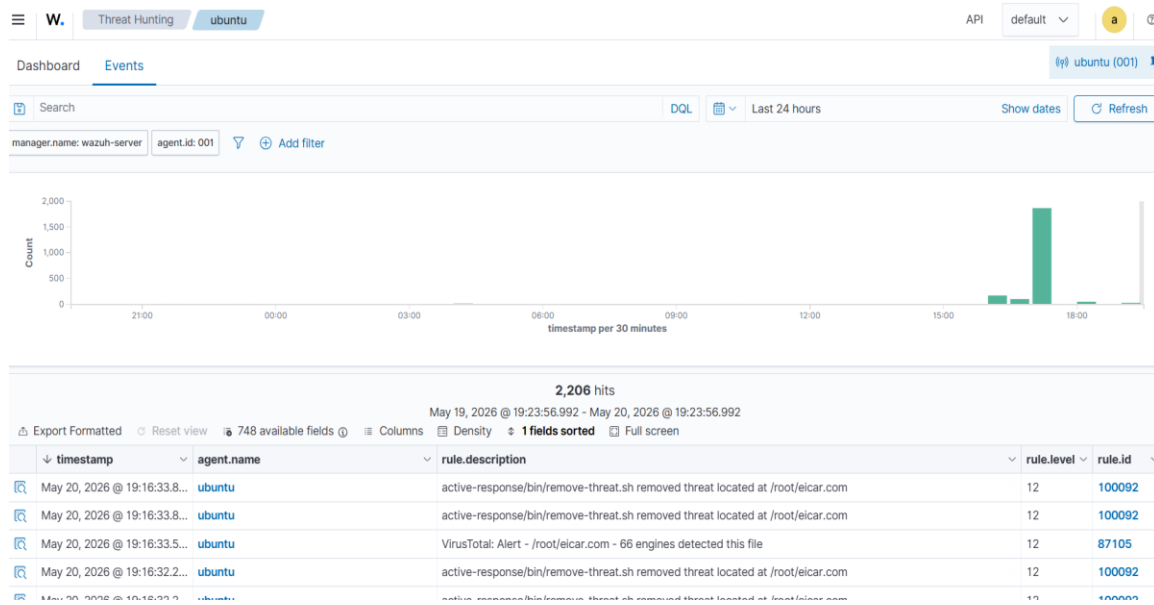


```
ubuntu@ubuntu-VirtualBox: ~  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
jq is already the newest version (1.7.1-3ubuntu0.24.04.2).  
The following package was automatically installed and is no longer required:  
  libwoff1  
Use 'sudo apt autoremove' to remove it.  
0 upgraded, 0 newly installed, 0 to remove and 185 not upgraded.  
ubuntu@ubuntu-VirtualBox:~$ sudo nano ^[[200~/var/ossec/active-response/bin/remove-threat.sh~  
ubuntu@ubuntu-VirtualBox:~$ sudo nano /var/ossec/active-response/bin/remove-threat.sh  
ubuntu@ubuntu-VirtualBox:~$ sudo chmod 750 /var/ossec/active-response/bin/remove-threat.sh  
sudo chown root:wazuh /var/ossec/active-response/bin/remove-threat.sh  
ubuntu@ubuntu-VirtualBox:~$ sudo systemctl restart wazuh-agent  
ubuntu@ubuntu-VirtualBox:~$ sudo curl -Lo /root/eicar.com https://secure.eicar.org/eicar.com && sudo ls -lah /root/eicar.com  
[sudo] password for ubuntu:  
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current  
000    68   100    68    0     0    10      0  0:00:06  0:00:06  --:--:--   15  
-rw-r--r--  1 root root 68 May 21 02:16 /root/eicar.com  
ubuntu@ubuntu-VirtualBox:~$
```

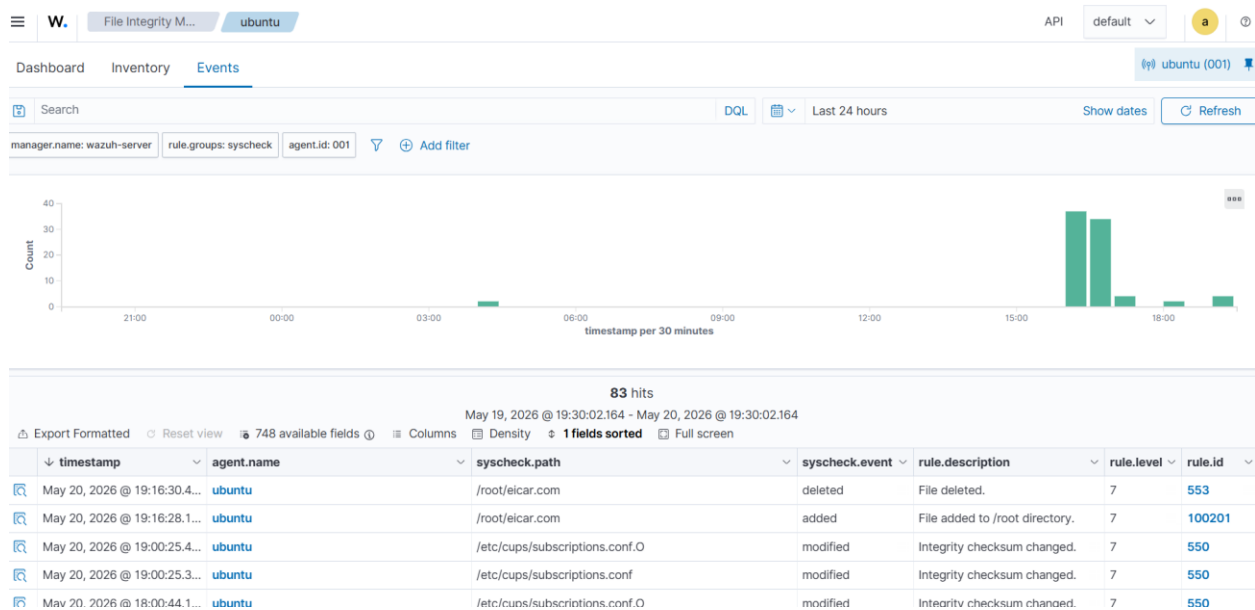
Visualising Alerts

Threat Hunting

Wazuh dashboard. To do this, go to the **Threat Hunting** module and add the filters in the search bar to query the alerts.



File Integrity Monitoring



API Monitoring

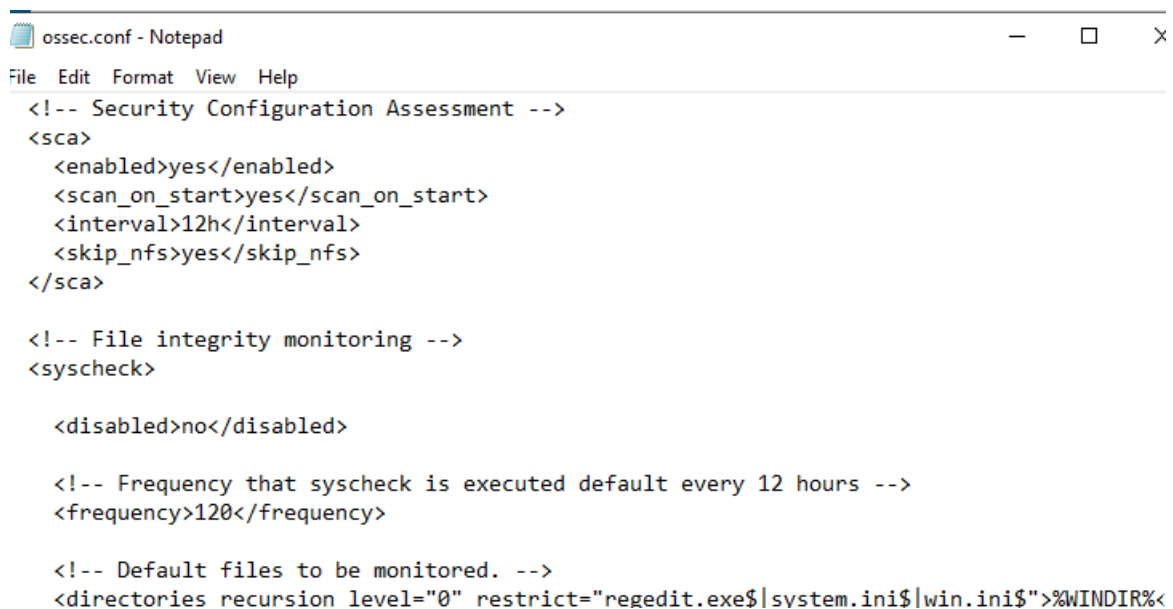


Configuration for the Windows endpoint

1. Search for syscheck using C:\Program Files (x86)\ossec-agent\ossec.conf

Notepad as administrator – file-open-local dis-progx86-ossec agent – change files to all files – ossec.conf – open

Ensured disabled was set to no and I changed my frequency from 43200 to 120

A screenshot of a Notepad window titled 'ossec.conf - Notepad'. The window has a menu bar with 'File', 'Edit', 'Format', 'View', and 'Help'. The text content is as follows:

```
<!-- Security Configuration Assessment -->
<sca>
  <enabled>yes</enabled>
  <scan_on_start>yes</scan_on_start>
  <interval>12h</interval>
  <skip_nfs>yes</skip_nfs>
</sca>

<!-- File integrity monitoring -->
<syscheck>

  <disabled>no</disabled>

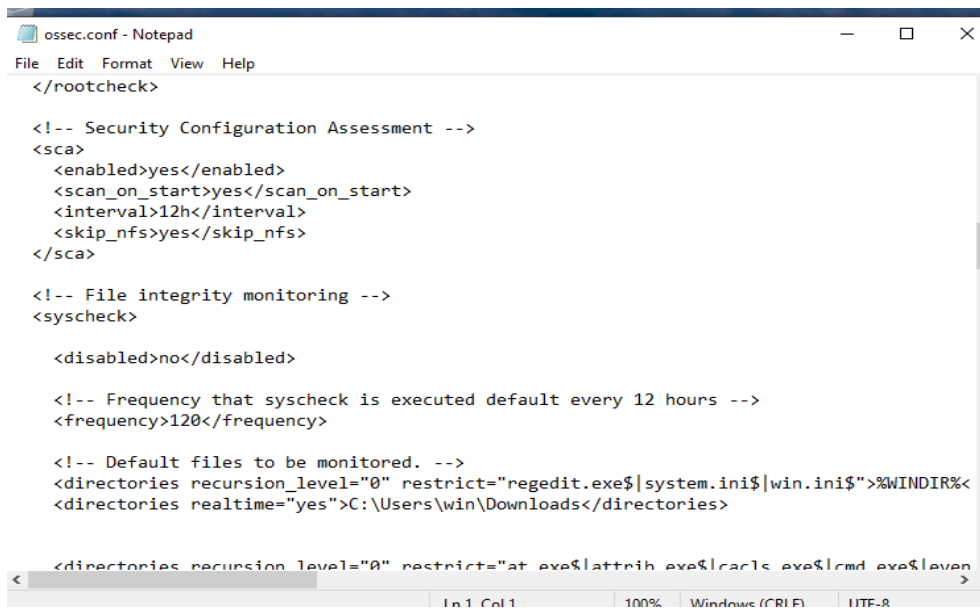
  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>120</frequency>

  <!-- Default files to be monitored. -->
  <directories recursion level="0" restrict="regedit.exe|system.ini|win.ini">%WINDIR%<
```

2. In the syscheck, configure wazuh to monitor C:\Users\<USER_NAME>\Downloads

```
<directories realtime="yes">C:\Users\<USER_NAME>\Downloads</directories>
```

USER_NAME in my case was replaced with win base on my path to Download files



```
</rootcheck>

<!-- Security Configuration Assessment -->
<sca>
  <enabled>yes</enabled>
  <scan_on_start>yes</scan_on_start>
  <interval>12h</interval>
  <skip_nfs>yes</skip_nfs>
</sca>

<!-- File integrity monitoring -->
<syscheck>

  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>120</frequency>

  <!-- Default files to be monitored. -->
  <directories recursion_level="0" restrict="regedit.exe|system.ini|win.ini">%WINDIR%<
  <directories realtime="yes">C:\Users\win\Downloads</directories>

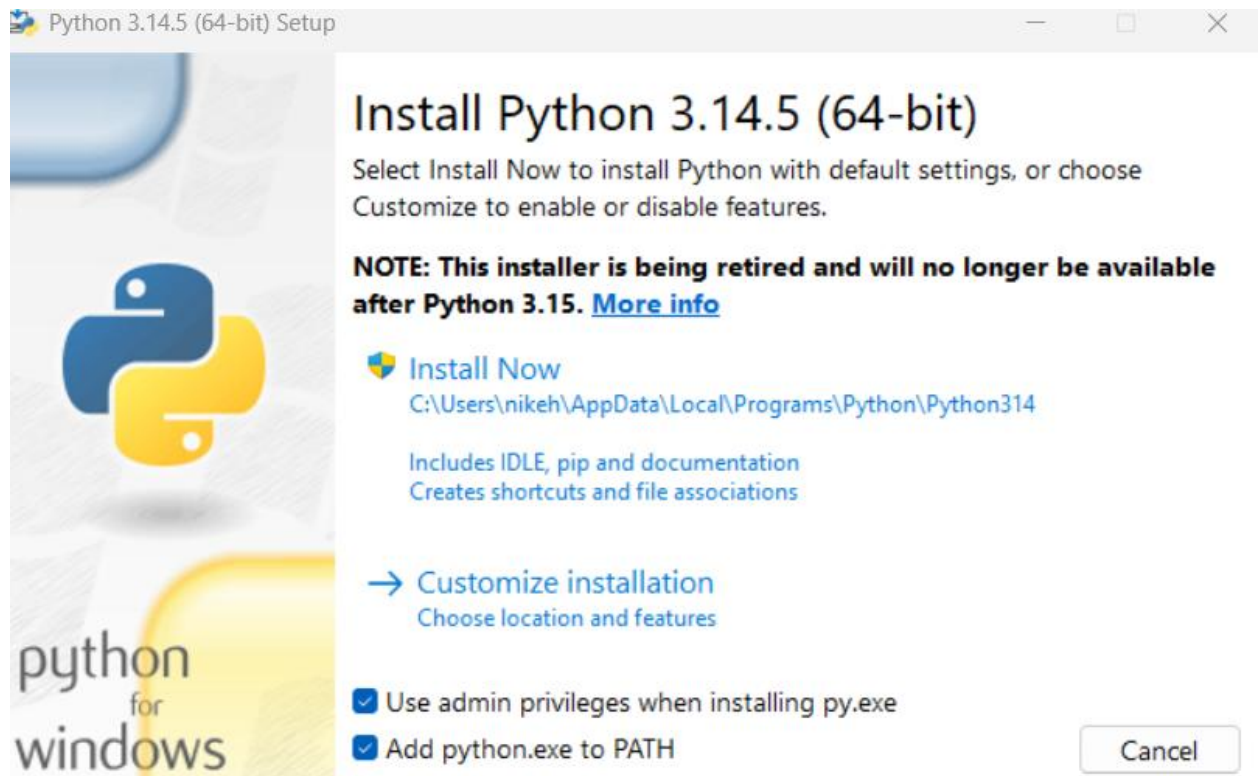
  <directories recursion_level="0" restrict="at.exe|attrib.exe|carls.exe|cmd.exe|even>
```

3. Download the Python executable installer from the [official Python website](#).

Downloaded [Windows installer \(64-bit\)](#)- stable release

4. Run the Python installer once downloaded. Make sure to check the following boxes:

- a. **Install launcher for all users**
- b. **Add Python 3.X to PATH** (This places the interpreter in the execution path)



5. Open windows PowerShell terminal and use pip to install PyInstaller:

```
> pip install pyinstaller
```

```
> pyinstaller --version
```

use Pyinstaller here to convert the active response Python script into an executable application that can run on a Windows endpoint.

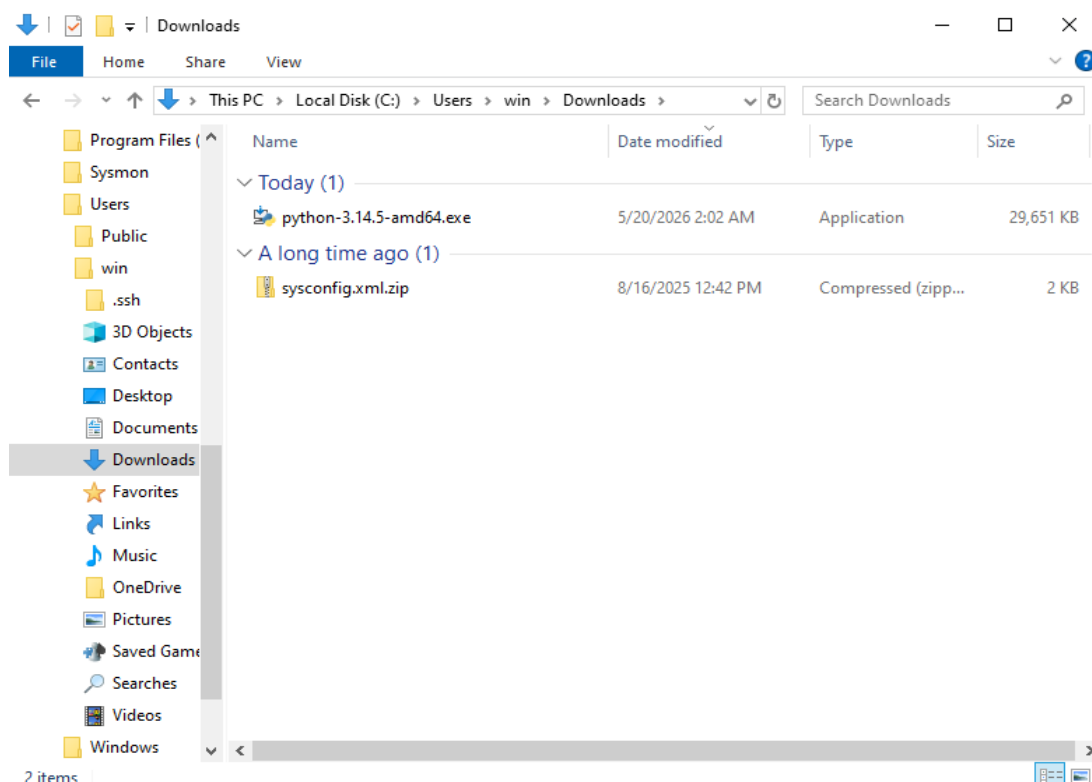
Got an error code stating

DEPRECATION: Running PyInstaller as admin is not necessary nor sensible. Run PyInstaller from a non-administrator terminal. PyInstaller 7.0 will block this. ERROR: Do not run pyinstaller from C:\Windows\system32. cd to where your code is and run pyinstaller from there.

Hint: You can open a terminal where your code is by going to the parent folder in Windows file explorer and typing cmd into the address bar. PS C:\Windows\system32>

Fixed it by following the hint given above

Go to the parent folder in Windows file explorer and typing cmd into the address bar. A terminal opens and ran `pyinstaller -version` which converted the python script to exe.



A terminal opens and ran `pyinstaller --version`. Python is converted to exe.

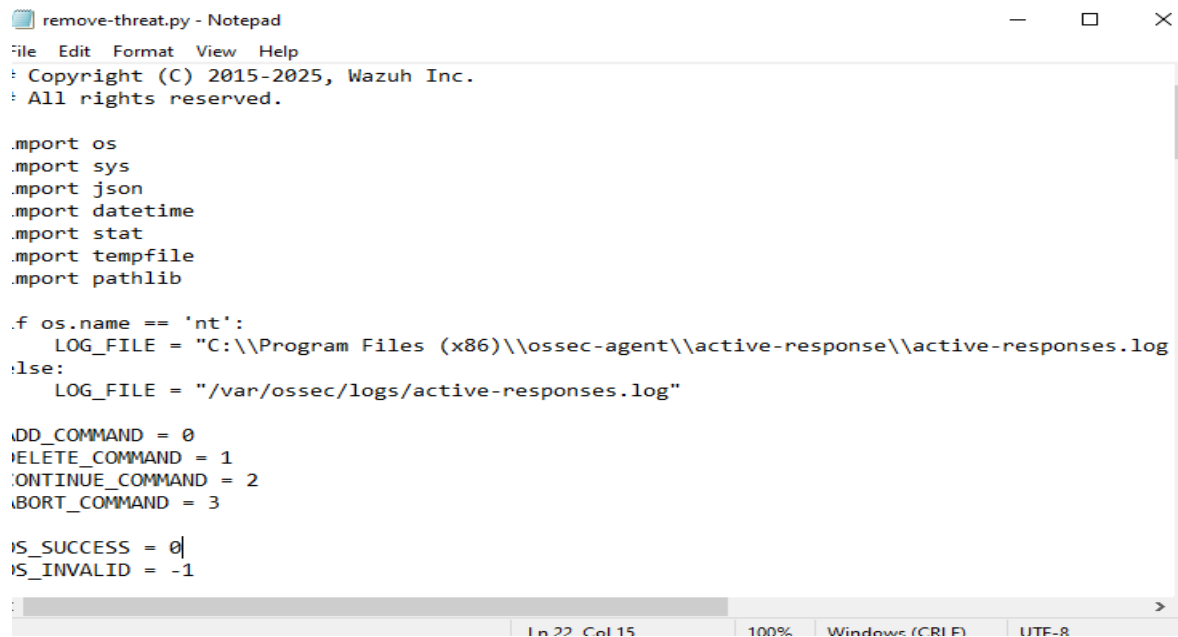
```
cmd: Select C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.6093]
(c) Microsoft Corporation. All rights reserved.

C:\Users\win\Downloads>pyinstaller --version
5.20.0

C:\Users\win\Downloads>
```

6. Create an active response script `remove-threat.py` to remove a file from the Windows endpoint:

I opened a notepad, the details in no.6 and saved it in the document as [remove-threat.py](#)



```
remove-threat.py - Notepad
File Edit Format View Help
: Copyright (C) 2015-2025, Wazuh Inc.
: All rights reserved.

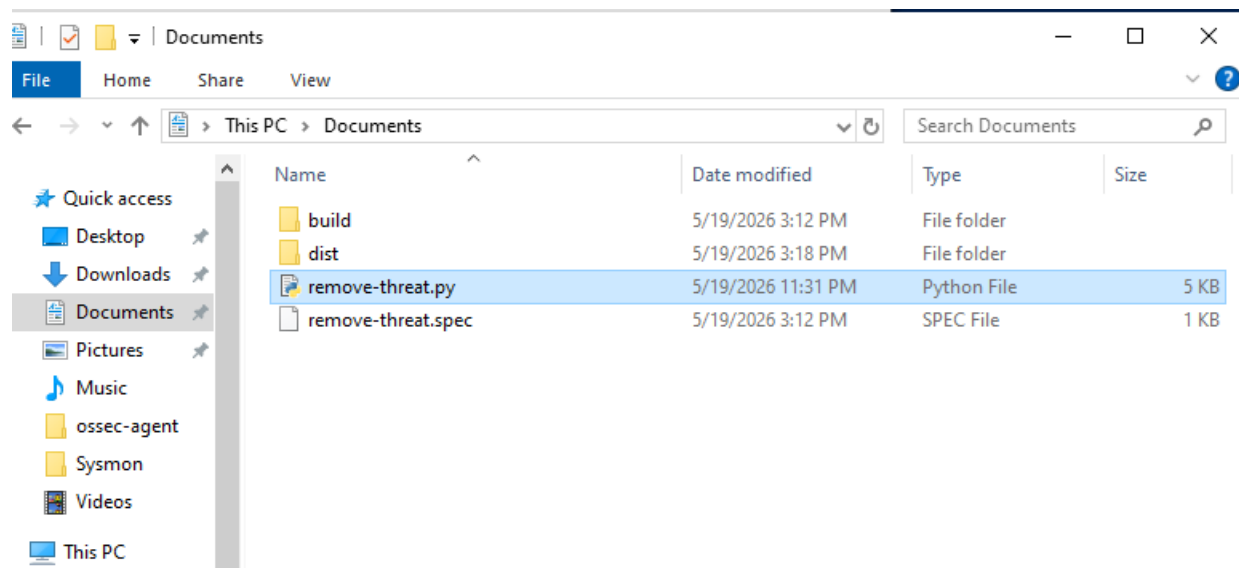
import os
import sys
import json
import datetime
import stat
import tempfile
import pathlib

if os.name == 'nt':
    LOG_FILE = "C:\\Program Files (x86)\\ossec-agent\\active-response\\active-responses.log"
else:
    LOG_FILE = "/var/ossec/logs/active-responses.log"

ADD_COMMAND = 0
DELETE_COMMAND = 1
CONTINUE_COMMAND = 2
ABORT_COMMAND = 3

IS_SUCCESS = 0
IS_INVALID = -1

Ln 22, Col 15    100%    Windows (CR LF)    UTF-8
```

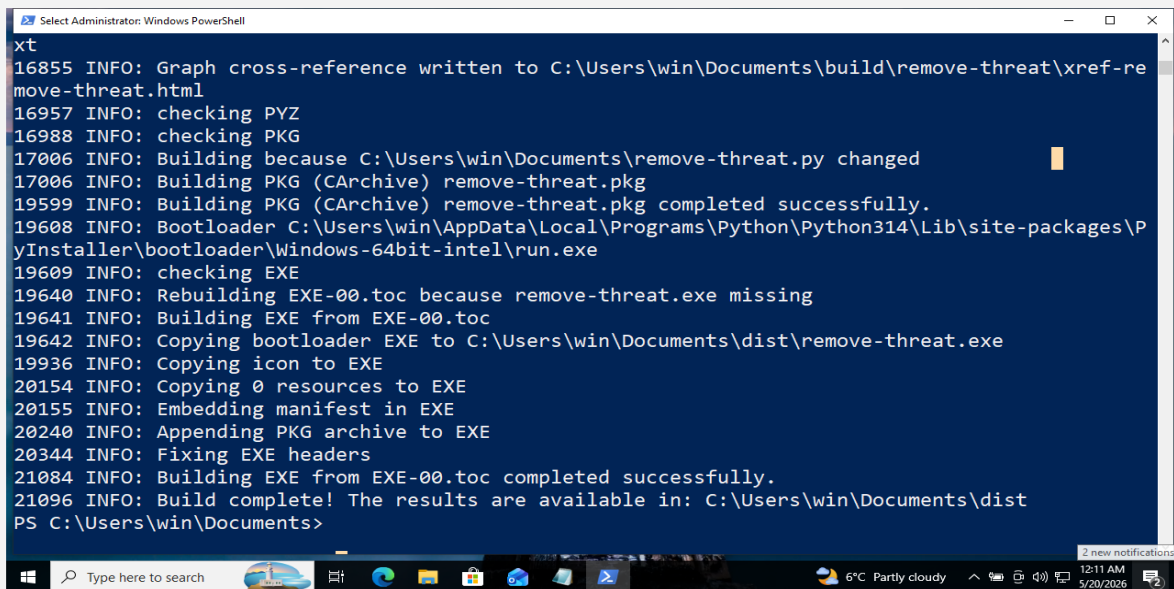


7. Converting the active response Python script `remove-threat.py` to a Windows executable application.

Since `pyinstaller` is given error on powershell. Error suggest running on cmd through parent folder

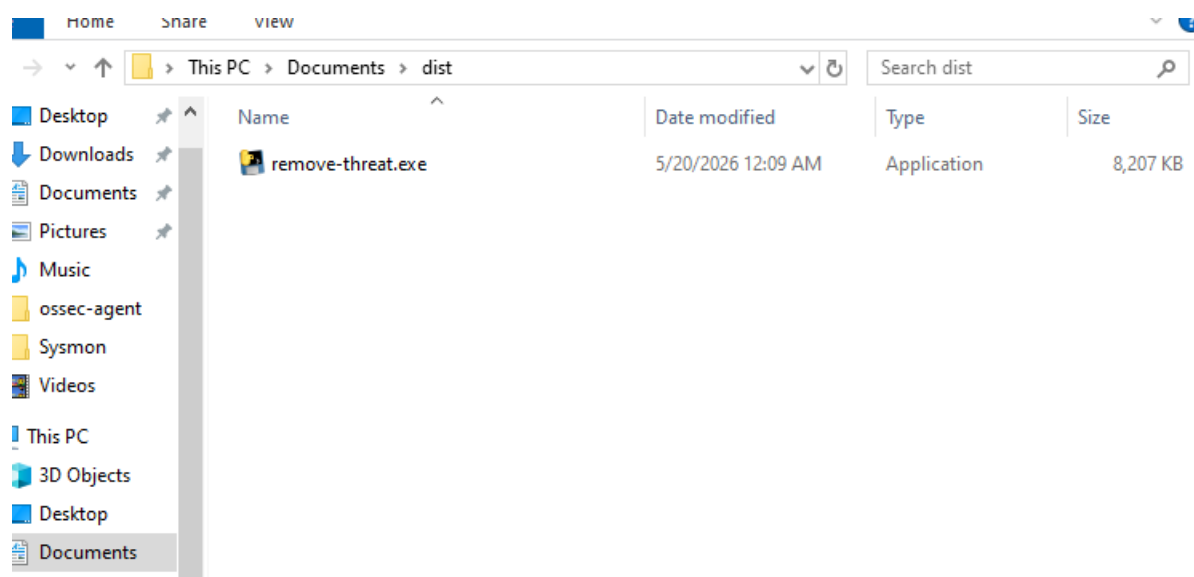
Path - `C:\Users\win\Documents`

Then run `pyinstaller -F C:\Users\win\Documents\remove-threat.py`

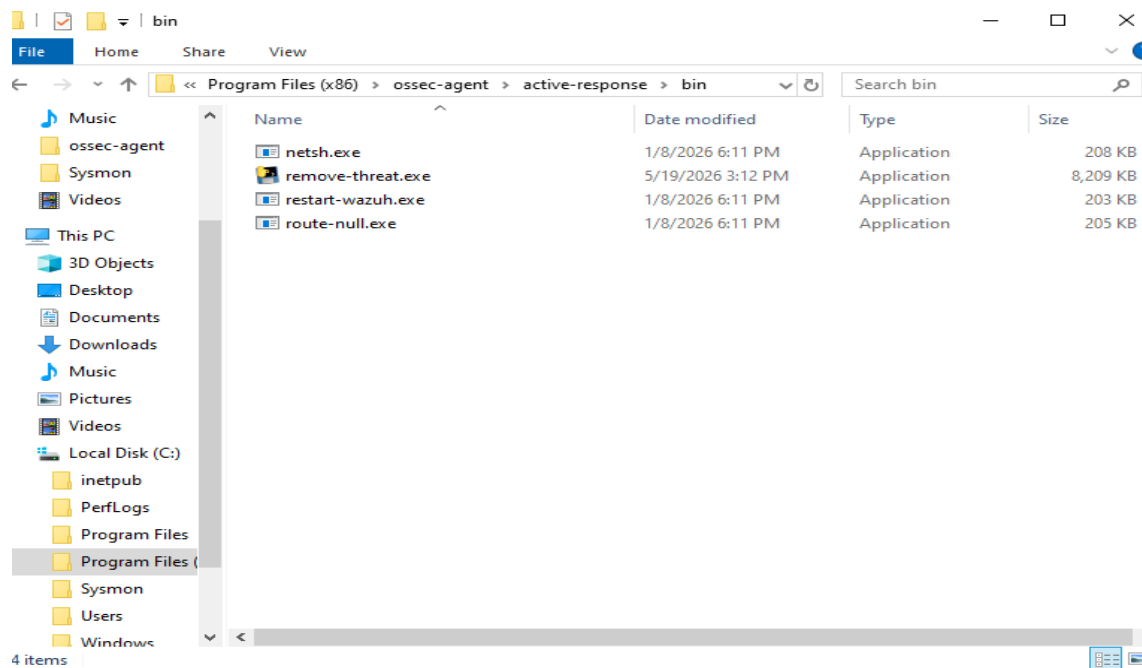


```
Select Administrator: Windows PowerShell

xt
16855 INFO: Graph cross-reference written to C:\Users\win\Documents\build\remove-threat\xref-re
move-threat.html
16957 INFO: checking PYZ
16988 INFO: checking PKG
17006 INFO: Building because C:\Users\win\Documents\remove-threat.py changed
17006 INFO: Building PKG (CArchive) remove-threat.pkg
19599 INFO: Building PKG (CArchive) remove-threat.pkg completed successfully.
19608 INFO: Bootloader C:\Users\win\AppData\Local\Programs\Python\Python314\Lib\site-packages\Py
Installer\bootloader\Windows-64bit-intel\run.exe
19609 INFO: checking EXE
19640 INFO: Rebuilding EXE-00.toc because remove-threat.exe missing
19641 INFO: Building EXE from EXE-00.toc
19642 INFO: Copying bootloader EXE to C:\Users\win\Documents\dist\remove-threat.exe
19936 INFO: Copying icon to EXE
20154 INFO: Copying 0 resources to EXE
20155 INFO: Embedding manifest in EXE
20240 INFO: Appending PKG archive to EXE
20344 INFO: Fixing EXE headers
21084 INFO: Building EXE from EXE-00.toc completed successfully.
21096 INFO: Build complete! The results are available in: C:\Users\win\Documents\dist
PS C:\Users\win\Documents>
```



8. Move the executable file remove-threat.exe to the C:\Program Files (x86)\ossec-agent\active-response\bin directory.



9. Restart-Service -Name wazuh

Wazuh Server

Same as Ubuntu, to activate wazuh on windows to enable typinghatgpt, typed "How do I confirm the distribution of my linux system"

`cat /etc/os-release`

`Sudo systemctl status sshd`

```

NAME="Amazon Linux"
VERSION="2023"
ID="amazon"
ID_LIKE="fedora"
VERSION_ID="2023"
PLATFORM_ID="platform:al2023"
PRETTY_NAME="Amazon Linux 2023.11.20260413"
ANSI_COLOR="0;33"
CPE_NAME="cpe:2.3:amazon:amazon_linux:2023"
HOME_URL="https://aws.amazon.com/linux/amazon-linux-2023/"
DOCUMENTATION_URL="https://docs.aws.amazon.com/linux/"
SUPPORT_URL="https://aws.amazon.com/premiumsupport/"
BUG_REPORT_URL="https://github.com/amazonlinux/amazon-linux-2023"
VENDOR_NAME="AWS"
VENDOR_URL="https://aws.amazon.com/"
SUPPORT_END="2029-06-30"

[wazuh-user@wazuh-server ~]$ sudo systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-05-17 00:07:49 UTC; 48min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 1996 (sshd)
      Tasks: 1 (limit: 4647)
     Memory: 2.6M
        CPU: 66ms
    CGroup: /system.slice/ssh.service
            └─1996 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

May 17 00:07:49 wazuh-server sshd[1996]: Server listening on 0.0.0.0 port 22.
May 17 00:07:49 wazuh-server sshd[1996]: Server listening on :: port 22.
[wazuh-user@wazuh-server ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:00:27:1f:15:42 brd ff:ff:ff:ff:ff:ff
    altname enp0s17
    inet 10.0.0.27/24 metric 1024 brd 10.0.0.255 scope global dynamic eth0
        valid_lft 169065sec preferred_lft 169065sec
    inet6 2604:3403:147c:31a0:aa02:128 scope global dynamic noprefixroute
        valid_lft 169065sec preferred_lft 169065sec
    inet6 2604:3403:147c:31a0:a0:27f1:f41f:1542/64 scope global dynamic mngtnpaddr noprefixroute
        valid_lft 172798sec preferred_lft 172798sec
    inet6 fe80::271f:1542:fe80::271f:1542/64 scope link link_proto kernel_ll
        valid_lft forever preferred_lft forever
[wazuh-user@wazuh-server ~]$

```

In Windows Powershell, type ssh wazuh-user@10.0.0.27 to enable wazuh

1. [/var/ossec/etc/ossec.conf](#) file on the Wazuh server to enable the VirusTotal integration. Replaced <YOUR_VIRUS_TOTAL_API_KEY> with your my [VirusTotal API key](#) from the virus total webpage.

```

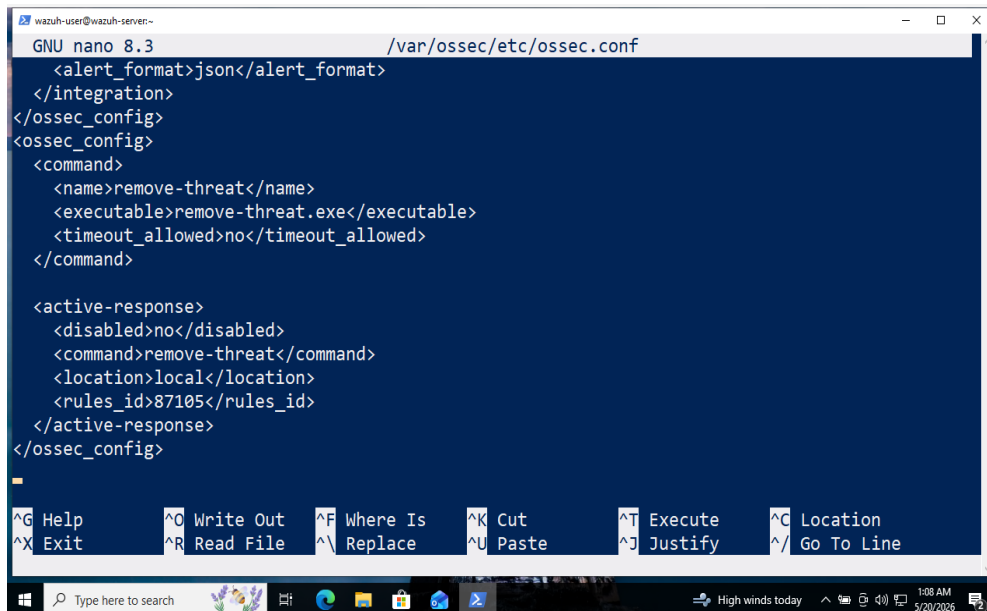
GNU nano 8.3 /var/ossec/etc/ossec.conf
</command>

<active-response>
  <disabled>no</disabled>
  <command>remove-threat</command>
  <location>local</location>
  <rules_id>87105</rules_id>
</active-response>
</ossec_config>
<ossec_config>
  <integration>
    <name>virustotal</name>
    <api_key>ab268c19855a4869bb6798732c2ee44cddb53aba299eff3974a45cf08c1973b0</api_key> <!-- R
    <group>syscheck</group>
    <alert_format>json</alert_format>
  </integration>
</ossec_config>
</ossec_config>

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^U Replace   ^U Paste     ^J Justify  ^_ Go To Line

```

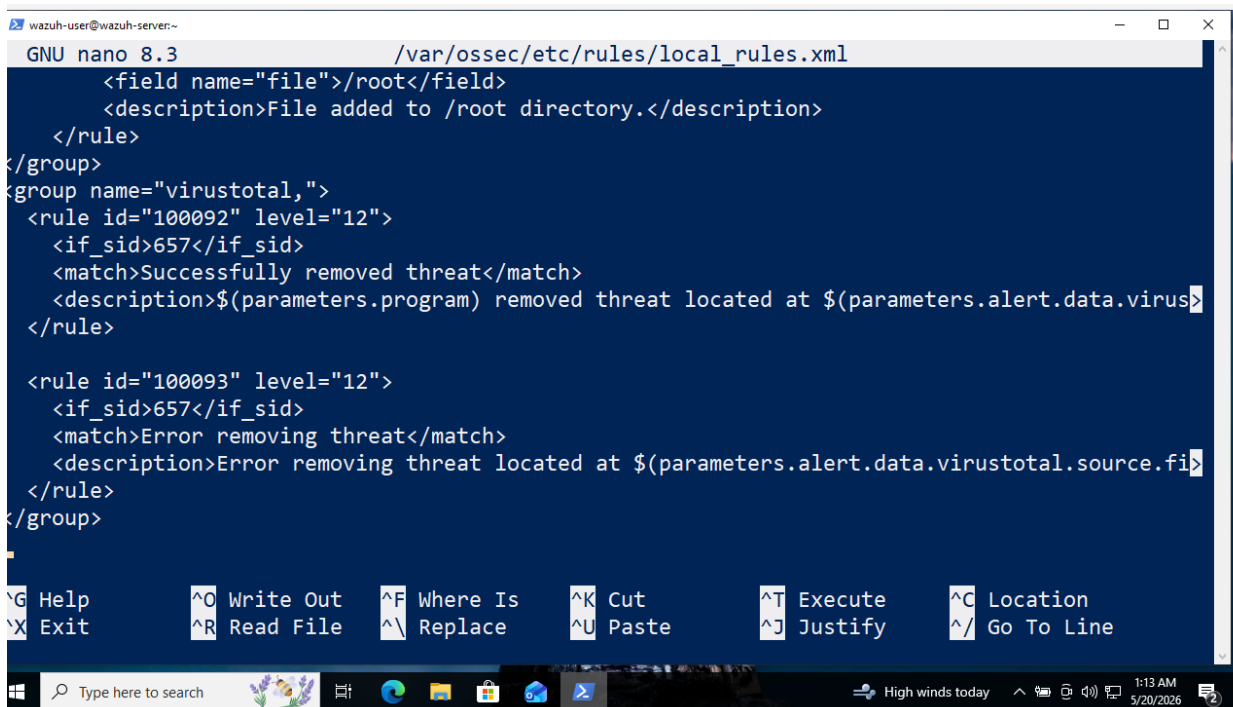
2.Enable active response by inserting the command below.



```
GNU nano 8.3 /var/ossec/etc/ossec.conf
<alert_format>json</alert_format>
</integration>
</ossec_config>
<ossec_config>
  <command>
    <name>remove-threat</name>
    <executable>remove-threat.exe</executable>
    <timeout_allowed>no</timeout_allowed>
  </command>

  <active-response>
    <disabled>no</disabled>
    <command>remove-threat</command>
    <location>local</location>
    <rules_id>87105</rules_id>
  </active-response>
</ossec_config>
```

3. To alert about active response /var/ossec/etc/rules/local_rules.xml, type the response result below



```
GNU nano 8.3 /var/ossec/etc/rules/local_rules.xml
  <field name="file">/root</field>
  <description>File added to /root directory.</description>
</rule>
</group>
<group name="virustotal,">
  <rule id="100092" level="12">
    <if_sid>657</if_sid>
    <match>Successfully removed threat</match>
    <description>$(parameters.program) removed threat located at $(parameters.alert.data.virus
  </rule>

  <rule id="100093" level="12">
    <if_sid>657</if_sid>
    <match>Error removing threat</match>
    <description>Error removing threat located at $(parameters.alert.data.virustotal.source.fi
  </rule>
</group>
```

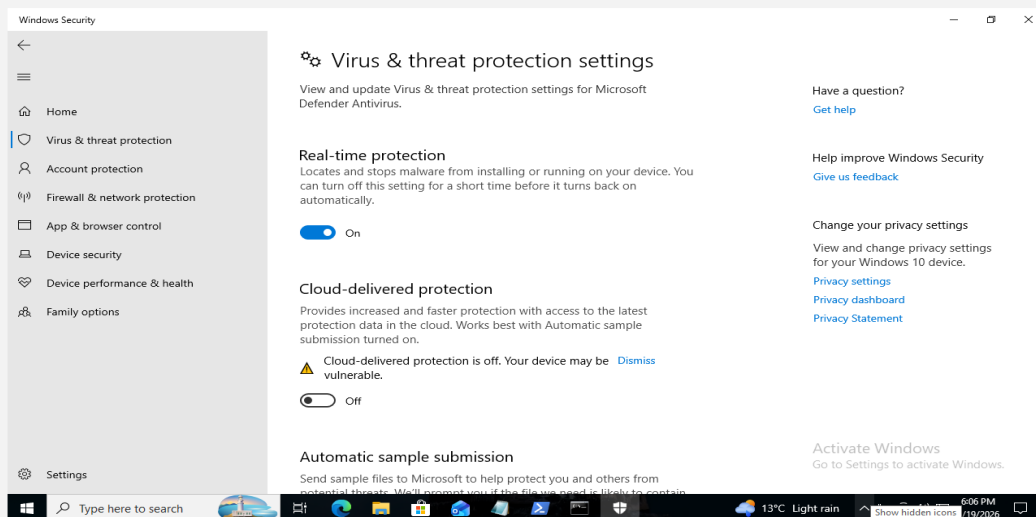
4.Restart for configuration changes

sudo systemctl restart wazuh-manager

```
wazuh-user@wazuh-server~$ cat /etc/network/interfaces
inet        /8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1f:15:42 brd ff:ff:ff:ff:ff:ff
    altname enp0s17
    inet      /24 metric 1024 brd                scope global dynamic eth0
        valid_lft 164659sec preferred_lft 164659sec
    inet6 2001:3000:1000:1000:1000:1000:1000:1000/128 scope global dynamic noprefixroute
        valid_lft 164661sec preferred_lft 164661sec
    inet6 2001:3000:1000:1000:1000:1000:1000:1000/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 172799sec preferred_lft 172799sec
    inet6 fe80::208:1fff:fe1f:1542/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[wazuh-user@wazuh-server ~]$ sudo nano /var/ossec/etc/ossec.conf
[wazuh-user@wazuh-server ~]$ sudo nano /var/ossec/etc/ossec.conf
[wazuh-user@wazuh-server ~]$ sudo nano /var/ossec/etc/rules/local_rules.xml
[wazuh-user@wazuh-server ~]$ [wazuh-user@wazuh-server ~]$ sudo systemctl restart wazuh-manager
[wazuh-user@wazuh-server ~]$
```

Attack Emulation

1. Temporarily turn off real-time Microsoft Defender antivirus protection in Windows Security:



2. Download an [EICAR test](#) file to the C:\Users\<USER_NAME>\Downloads directory on the Windows endpoint, not the one embedded with wazuh

```
Windows PowerShell
Copyright (c) Microsoft Corporation. All rights reserved.

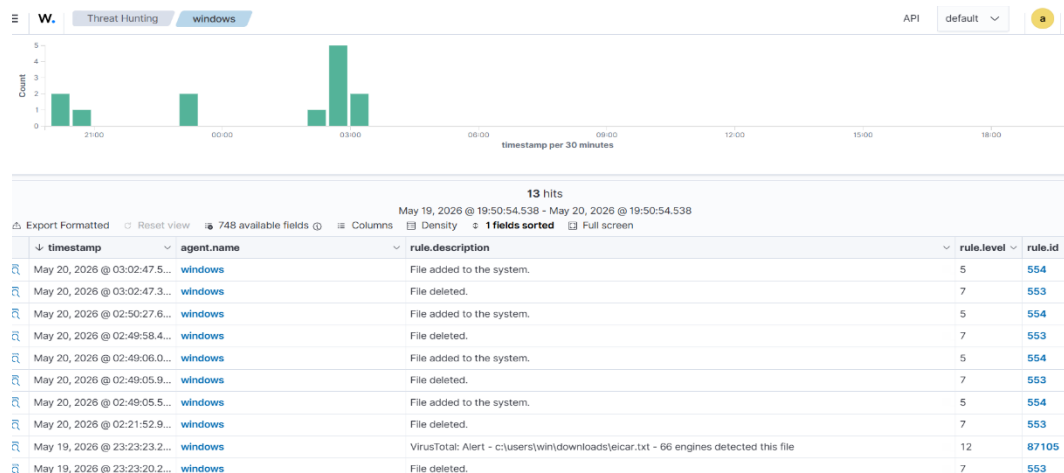
Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\win> Invoke-WebRequest -Uri https://secure.eicar.org/eicar.com.txt -OutFile eicar.txt
PS C:\Users\win> cp .\eicar.txt C:\Users\win\Downloads
PS C:\Users\win>
```

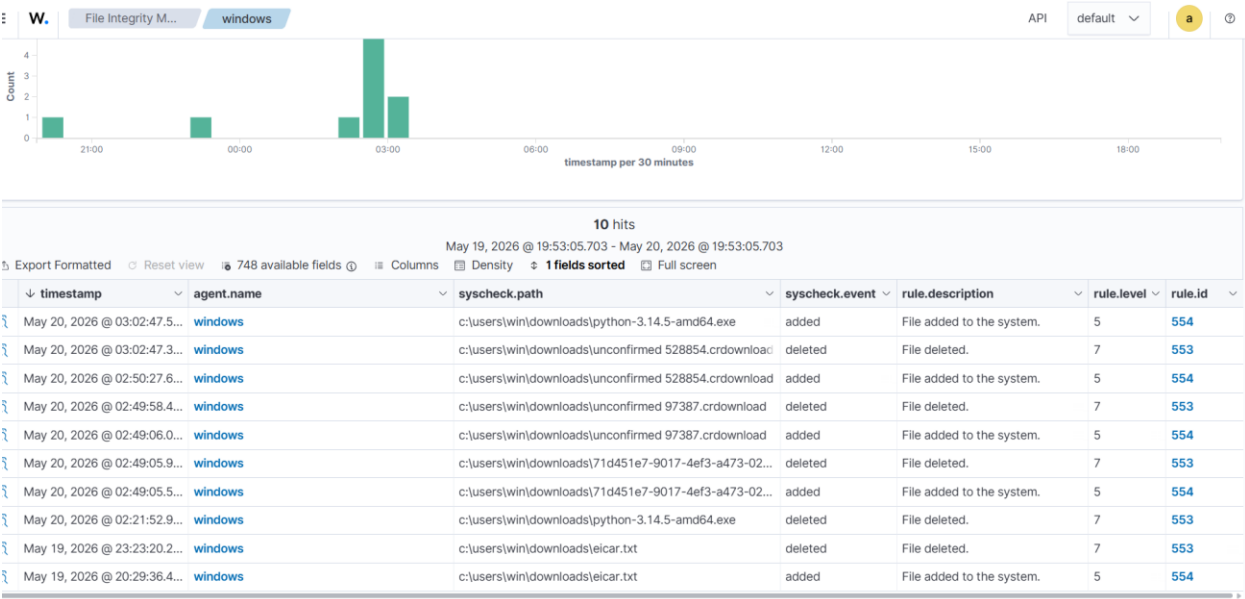
Visualise alert

- Visualising data on the wazuh dashboard through **Threat Hunting**

Added Windows - rule.id: is one of 554,100092,553,87105 in the filter to query alert



File integrity Monitoring



API Monitoring



